

Intermediate SOC Investigation Using Public Attack Data in Splunk

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Platform: Splunk Enterprise

Data Source: Windows Sysmon XML Event Logs

Dataset: Splunk Attack Data – First Time Windows Service

Investigation Date: September 2026

Observed Event Date: 9 December 2020

1. Project Overview

This project documents an intermediate Security Operations Center (SOC) investigation using a public Windows attack dataset in Splunk.

The investigation began with a broad review of imported Sysmon telemetry and progressively narrowed into suspicious PowerShell activity, Windows service creation, registry changes, file creation, and SYSTEM-level process execution.

The most significant finding was a simulated **unquoted service-path / path-interception scenario**. A Windows service was configured with the unquoted path:

`C:\Program Files\windows_service.exe`

The investigation later confirmed that Windows executed:

`C:\Program.exe`

through `services.exe` under the NT `AUTHORITY\SYSTEM` account.

The dataset also contained explicit Atomic Red Team references to T1574.009, supporting the assessment that the activity represents controlled attack-simulation behavior rather than confirmed real-world malware.

2. Investigation Objectives

The objectives of this investigation were to:

- validate successful ingestion of the attack dataset into Splunk;
- identify suspicious Windows process activity;
- investigate repeated PowerShell `EncodedCommand` execution;
- reconstruct relevant parent-child process relationships;
- investigate Windows service creation and registry modifications;
- determine how `C:\Program.exe` was created and executed;
- reconstruct a consolidated activity timeline;

- map observed behaviors to MITRE ATT&CK;
- develop a reusable Splunk detection for similar service-path behavior.

3. Dataset and Environment

The Sysmon dataset was imported into the Splunk index:

`attack_data`

with sourcetype:

`XmlWinEventLog`

The initial validation returned **10,290 events**.

The Splunk upload host was KALIFA, while the Windows system represented inside the attack logs was:

`win-dc-533.attackrange.local`

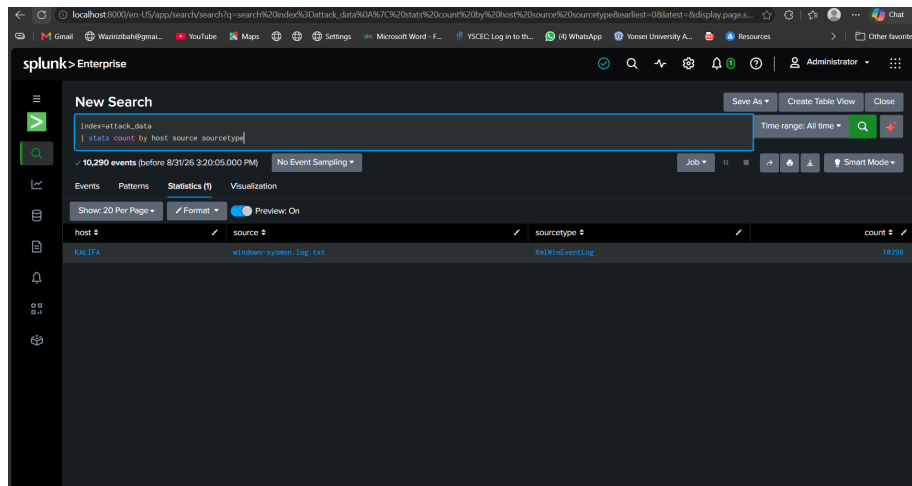


Figure 1: Attack data import confirmation

The investigation focused primarily on Sysmon process creation, file creation, registry modification, and process access events.

4. Investigation Methodology

The investigation followed a SOC-style workflow:

1. Confirm data ingestion and identify available Sysmon Event IDs.

2. Review process creation activity.
3. Classify common and suspicious process behavior.
4. Isolate PowerShell executions using **EncodedCommand**.
5. Extract and decode relevant encoded content.
6. Review repeated execution patterns and process relationships.
7. Build a process activity timeline.
8. Investigate Windows service registry activity.
9. Trace suspicious service executable creation and execution.
10. Confirm SYSTEM-level execution.
11. Reconstruct the final activity timeline.
12. Map evidence to MITRE ATT&CK.
13. Convert the investigation finding into a reusable Splunk detection.

The investigation used the principle:

Evidence → Interpretation → Confidence → Recommendation

Suspicious behavior was not automatically treated as malicious. Conclusions were based on correlated evidence from multiple Sysmon event types.

5. Key Investigation Findings

5.1 Suspicious Encoded PowerShell Activity

Process-creation analysis identified repeated PowerShell execution using the **EncodedCommand** parameter.

A total of **28 encoded PowerShell events** were identified for:

ATTACKRANGE\Administrator

on:

win-dc-533.attackrange.local

The activity involved parent-child relationships including:

- **cmd.exe -> powershell.exe**
- **powershell.exe -> powershell.exe**

Repeated encoded execution is security-relevant because it may reduce the readability of commands during routine monitoring. However, encoded PowerShell alone is not sufficient evidence of malicious intent.

5.2 Encoded Command Decoding

The investigation extracted encoded command content and decoded selected PowerShell payloads.

security-relevant. At this stage, however, the activity was still treated as suspicious rather than automatically classified as malicious.

5.3 Windows Service Registry Investigation

The investigation later examined registry activity under the Windows Services registry path.

A broad search initially returned a large number of service-related events, so the search was narrowed to service `ImagePath` modifications.

Two relevant `ImagePath` entries were identified:

- **npf**, associated with the Splunk Universal Forwarder environment;
- **Example Service**, configured to execute:

C:\Program Files\windows_service.exe

[illegible]

Figure 4: Service ImagePath registry changes

The **npf** entry was consistent with Splunk-related activity and was treated as low priority for this investigation.

Example Service was more significant because its executable path contained a space and was configured without quotation marks.

5.4 Service Creation and Execution Sequence

Further investigation correlated the suspicious service configuration with file and process activity.

The following sequence was observed:

1. WindowsServiceExample.exe was copied toward the service-path test.
2. C:\Program Files\windows_service.exe was created.
3. The Example Service ImagePath was configured.
4. sc.exe created Example Service.
5. sc.exe started the service.
6. A process executed through services.exe.
7. The service was stopped.
8. The service was deleted.

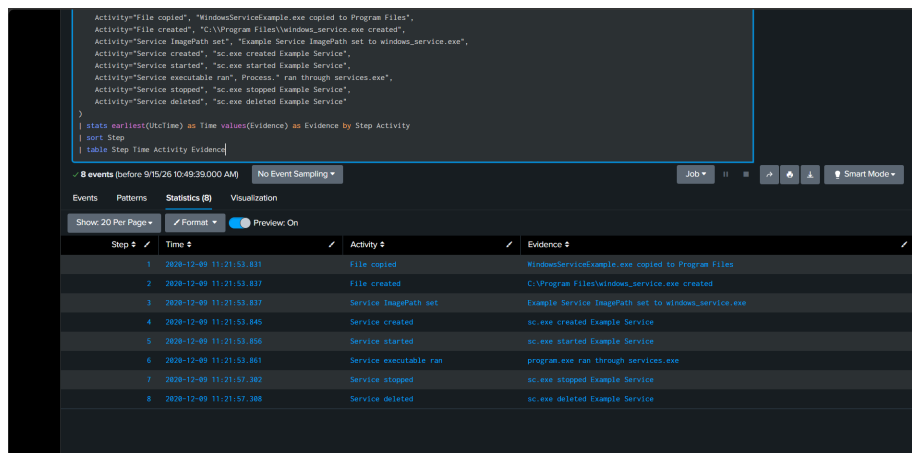


Figure 5: Windows service investigation summary

The underlying command data also referenced:

C:\AtomicRedTeam\atomics\T1574.009\bin\WindowsServiceExample.exe

This was a major contextual indicator that the dataset represented Atomic Red Team attack simulation.

5.5 SYSTEM-Level Execution of C:\Program.exe

A focused process-creation search identified the strongest execution evidence in the investigation.

Sysmon recorded:

- **Process:** program.exe
- **Image:** C:\Program.exe
- **Parent process:** services.exe
- **User:** NT AUTHORITY\SYSTEM
- **Command line:** C:\Program Files\windows_service.exe

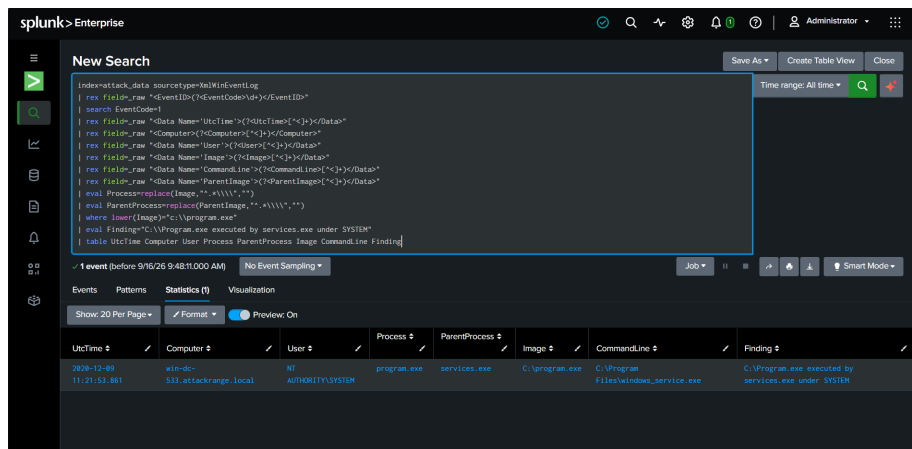


Figure 6: Program.exe execution

The difference between the actual process image and the command line was significant.

The service referenced:

C:\Program Files\windows_service.exe

but the process Windows actually launched was:

C:\Program.exe

This behavior is consistent with path interception involving an unquoted executable path containing spaces.

5.6 Origin of C:\Program.exe

The investigation then traced how C:\Program.exe appeared on the host.

The logs showed:

- a command referencing the Atomic Red Team WindowsServiceExample.exe;
- creation of C:\Program.exe;
- creation and startup of Example Service;
- process-access activity involving C:\Program.exe;
- execution of C:\Program.exe under SYSTEM;
- subsequent service stop and deletion activity.

This closed an important evidence gap. The investigation did not merely observe C:\Program.exe executing; it also established its creation immediately before the service execution sequence.

6. Consolidated Activity Timeline

The final timeline consolidated the strongest findings from the earlier investigation stages.

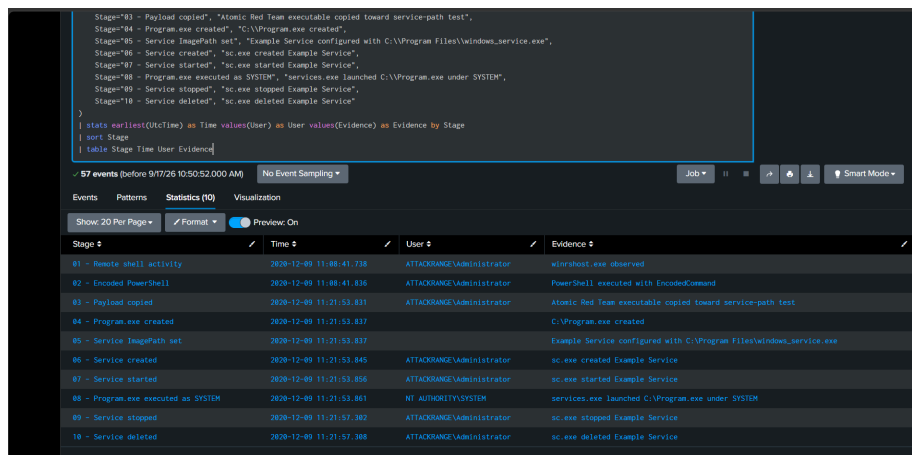


Figure 7: Final attack timeline

Stage	Observed Activity
1	Remote shell activity through <code>winrmshost.exe</code>
2	PowerShell execution using <code>EncodedCommand</code>
3	Atomic Red Team executable copied
4	<code>C:\Program.exe</code> created
5	Example Service ImagePath configured
6	Example Service created
7	Example Service started
8	<code>C:\Program.exe</code> executed by <code>services.exe</code> under SYSTEM
9	Example Service stopped
10	Example Service deleted

The earlier remote-shell and encoded-PowerShell activity occurred on the same system and account context before the later service-path activity. The timeline therefore presents them as related observed activity. It does **not** claim that every earlier event directly caused every later event unless supported by the available logs.

7. MITRE ATT&CK Mapping

The following ATT&CK techniques are supported by the observed evidence.

Observed Behavior	MITRE ATT&CK
Unquoted path resulted in C:\Program.exe execution	T1574.009 – Path Interception by Unquoted Path
PowerShell executed with EncodedCommand	T1059.001 – PowerShell
cmd.exe used for command execution	T1059.003 – Windows Command Shell
sc.exe created and started a Windows service	T1543.003 – Windows Service
whoami.exe used for account/user discovery	T1033 – System Owner/User Discovery

Primary Technique

The primary technique is:

T1574.009 – Path Interception by Unquoted Path

The evidence supporting this mapping includes:

- an unquoted service path containing a space;
- creation of C:\Program.exe;
- service startup;
- execution of C:\Program.exe by services.exe;
- SYSTEM-level execution;
- explicit Atomic Red Team T1574.009 references in the dataset.

MITRE describes this behavior as taking advantage of file paths that lack surrounding quotation marks, allowing Windows to resolve and execute an unintended executable from a higher-level directory.

8. Detection Engineering

After completing the investigation, a reusable Splunk detection was developed.

The rule searches Sysmon process-creation events for executables launched by **services.exe** directly from the root of a drive.

The detection does not hard-code C:\Program.exe. Instead, it looks for a broader pattern such as:

- parent process = **services.exe**;
- executable launched directly from a drive root;

- In the dataset, the detection returned one high-priority result:

- services.exe -> C:\Program.exe
- user: NT AUTHORITY\SYSTEM
- command line: C:\Program Files\windows_service.exe

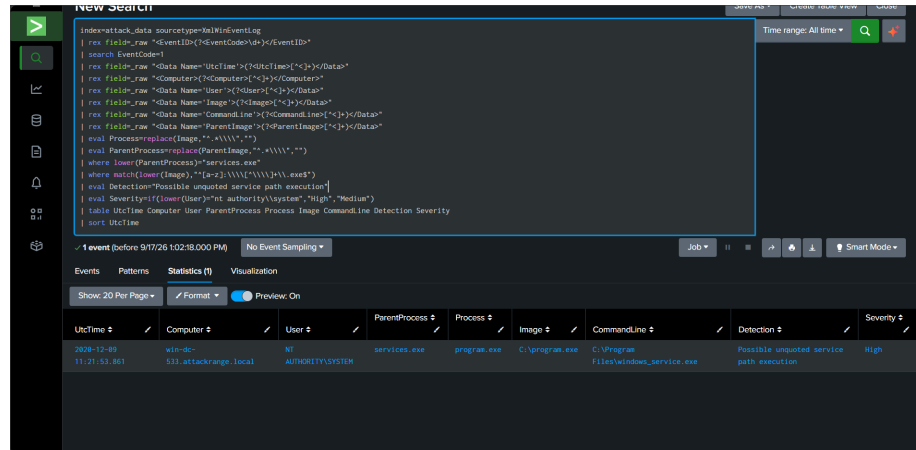


Figure 8: Unquoted service path detection

The full SPL is stored separately in:

splunk_queries/19_unquoted_service_path_detection.spl

The **Severity** and **Detection** fields shown in the result are analyst-created fields used to make the detection output easier to interpret. They are not native Sysmon fields.

9. Analyst Assessment

Evidence

The investigation confirmed:

- repeated encoded PowerShell execution;
- remote shell and command-shell activity;
- creation of **Example Service**;
- an unquoted service executable path;
- Atomic Red Team T1574.009 references;
- creation of C:\Program.exe;
- execution of C:\Program.exe by services.exe;
- execution under NT AUTHORITY\SYSTEM;

- service cleanup shortly afterwards.

Interpretation

The combined evidence is consistent with a controlled simulation of **Path Interception by Unquoted Path** involving a Windows service.

The strongest technical indicator is the mismatch between:

Configured/command-line path

C:\Program Files\windows_service.exe

and:

Actual executable image

C:\Program.exe

with `services.exe` as the parent process and SYSTEM as the execution context.

Confidence

High confidence that the service-path activity represents simulated T1574.009 behavior.

The confidence is based on direct Sysmon process evidence, registry evidence, file creation evidence, service-control commands, and explicit Atomic Red Team references.

The earlier remote-shell and encoded-PowerShell activity is strongly associated with the same investigation context, but the available logs should not be used to claim an unproven causal relationship between every individual event.

10. Recommendations

For a production Windows environment, recommended defensive actions include:

- ensure executable service paths containing spaces are surrounded by quotation marks;
- audit Windows service `ImagePath` registry values for insecure path configurations;
- restrict write permissions to high-risk locations such as the root of C:\;
- monitor new executable creation in drive-root directories;
- alert when `services.exe` launches unusual executables from unexpected locations;
- monitor service creation and modification using `sc.exe`;

- monitor PowerShell `EncodedCommand` activity and correlate it with surrounding process behavior;
- correlate service registry changes, file creation, and process execution rather than relying on a single event;
- use application-control technologies where appropriate to restrict unauthorized executable placement and execution.

11. Scope and Limitations

This project used a **public attack-simulation dataset** rather than telemetry from a live production incident.

Therefore:

- the findings demonstrate investigation and detection methodology;
- they should not be interpreted as evidence of a real compromise;
- some activity may exist specifically to demonstrate ATT&CK techniques;
- conclusions are limited to the events available in the supplied dataset.

The project intentionally distinguishes between suspicious behavior, confirmed log evidence, and analyst interpretation.

12. Conclusion

This investigation progressed from broad Sysmon analysis to a focused reconstruction of simulated Windows service-path exploitation.

The project demonstrated the ability to:

- ingest and validate Windows telemetry in Splunk;
- extract fields from raw Sysmon XML;
- analyze parent-child process relationships;
- investigate encoded PowerShell;
- distinguish likely benign activity from security-relevant activity;
- correlate process, file, registry, and service events;
- reconstruct a chronological attack timeline;
- map evidence to MITRE ATT&CK;
- develop a reusable Splunk detection from an investigation finding.

The central finding was the execution of `C:\Program.exe` by `services.exe` under NT `AUTHORITY\SYSTEM` while the associated service referenced the unquoted path `C:\Program Files\windows_service.exe`.

In the context of the Atomic Red Team references present in the dataset, the evidence is consistent with a controlled simulation of **MITRE ATT&CK T1574.009 – Path Interception by Unquoted Path**.

13. Repository Structure

```
splunk-public-attack-dataset-investigation/
|-- README.md
|-- investigation_report.md
|-- reports/
|   `-- 18_mitre_attack_mapping.md
|-- screenshots/
|   |-- 01_attack_data_import_confirmed.png
|   |-- 05_suspicious_powershell_encoded_commands.png
|   |-- 07_decoded_powershell_wrapper_command.png
|   |-- 13b_service_imagepath_registry_changes.png
|   |-- 14_windows_service_exe_investigation_summary.png
|   |-- 15_program_exe_service_execution.png
|   |-- 17_final_attack_timeline.png
|   `-- 19_unquoted_service_path_detection.png
|-- splunk_queries/
|   `-- ...
`-- dataset/
    `-- ...
```

The complete `screenshots/` and `splunk_queries/` directories may contain additional investigation artifacts that were intentionally not embedded in this report to avoid repetition.

14. References

- MITRE ATT&CK – T1574.009, Path Interception by Unquoted Path: <https://attack.mitre.org/techniques/T1574/009/>
- MITRE ATT&CK – T1543.003, Windows Service: <https://attack.mitre.org/techniques/T1543/003/>
- MITRE ATT&CK – T1059.001, PowerShell: <https://attack.mitre.org/techniques/T1059/001/>
- MITRE ATT&CK – T1059.003, Windows Command Shell: <https://attack.mitre.org/techniques/T1059/003/>
- MITRE ATT&CK – T1033, System Owner/User Discovery: <https://attack.mitre.org/techniques/T1033/>
- Splunk Attack Data dataset used in this project: https://github.com/splunk/attack_data